Aatman Mantri

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Summary

Senior Software Engineer with 5+ years of experience building and operating production backend systems in healthcare. I design multi-service architectures in Go, optimize high-traffic Django APIs (48 queries → 1), and manage Kubernetes infrastructure across GCP and Azure. Currently owning the full backend stack for medical imaging, authentication, and AI annotation platforms serving 4,100+ users.

Education

New York University Tandon School of Engineering — New York, USA
Master of Science in Computer Science; CGPA: 3.6/4 — *June 2023*

RV College of Engineering — Bengaluru, India
Bachelor of Engineering in Computer Science & Engineering; CGPA: 8.1/10 — *June 2019*

Experience

Senior Software Engineer — Oct 2023 – Present
Nference, Inc. — *Bengaluru, India*

- Designed and built a multi-service DICOM server from scratch in Go (API gateway, DICOMweb server, pathology service, LRU cleanup) with hot/cold storage orchestration, making 100M+ medical images accessible on-demand where only 1M were before. Deployed across 3 production environments (GCP + Azure).
- Rewrote the platform's most critical Django/DRF API endpoints — hit on every page load by 4,100+ users — reducing database queries from 48 to 11 (cold) / 1 (warm) via phased batching, BoolOr aggregation, polymorphic bypass, and a three-tier cache (request → Redis → DB).
- Simplified the AI Studio architecture from 5 data stores to 2 by removing ETCD, RabbitMQ, and BadgerDB (dead code and overengineering). Removed deprecated v1 APIs and 3 dead services from the codebase.
- Built end-to-end self-service signup flow with reCAPTCHA v3, admin approval, duration-based access expiry, and a transactional email system (10/10 mail-tester score). Shipped by business deadline for customer event.
- Turned an experimental DZI-to-DICOM conversion script into a production FastAPI service for pyramidal pathology image generation, integrated with the full imaging pipeline. Now used in customer sales pitches.
- Implemented zero-downtime blue-green deployments across Kubernetes environments, eliminating ~3 min of downtime per release. Diagnosed and fixed recurring pod OOM crashes traced to zombie slab memory from exec liveness probes.

Software Engineering Intern — June 2022 – Dec 2022
Nference, Inc. — *Boston, USA*

- Conceptualised and implemented a modified weighted PageRank algorithm to improve the ranking order of search results.
- Examined 3.9TB patient data for Amyloidosis pre-markers, halving doctors' preliminary analysis time through efficient insights.

Software Engineer — July 2019 – July 2021
Nference, Inc. — *Bengaluru, India*

- Built a distributed search engine serving queries over 30TB of medical data in 200ms with active-backup failover and automatic index updates.
- Engineered a distributed vector retrieval system using Intel MKL/CBLAS, achieving 100ms lookups across 3M 300-dimensional vectors.
- Managed multi-cloud deployments (Azure, GCP) for pharma clients under IRB compliance, maintaining 99.9% availability.

Technical Skills

Languages: Go, Python, SQL, C
Frameworks: Django/DRF, FastAPI, Gin
Infrastructure: Kubernetes, Helm, Docker, GCP, Azure, GitHub Actions
Databases: PostgreSQL, MongoDB, Redis
Tools: Grafana, Kafka, gRPC, Git, Orthanc/DICOM, MLFlow
Publication: Acute Coronary Syndrome Risk Prediction Model — IEEE Xplore